

CHAPTER 3: COMPUTER HARDWARE INPUT, PROCESS, OUTPUT, STORAGE AND NETWORK



INTENDED LEARNING OUTCOMES (ILOS)

AT THE END OF THIS LESSON, THE STUDENT IS EXPECTED TO:

- ❑ UNDERSTAND HOW THE COMPUTER HARDWARE WORKS;
- ❑ DIFFERENTIATE THE PURPOSE OF INPUT, PROCESS, OUTPUT, AND STORAGE AND NETWORK DEVICES;
- ❑ ENUMERATE ALL THE COMPUTER HARDWARE;
- ❑ DIFFERENCIATE THE TYPES OF COMPUTER MONITORS;
- ❑ UNDERSTAND HOW THE OUTPUT DEVICES WORKS;
- ❑ HAVE A GENERAL OVERVIEW ON PROCESSING DEVICES;
- ❑ EXPLAIN AND JUSTIFY HOW THE NETWORK DEVICES WORKS;
- ❑ LIST THE DIFFERENT TYPES OF HUB;
- ❑ LIST THE DIFFERENT TYPES OF BRIDGES

Computer Hardware

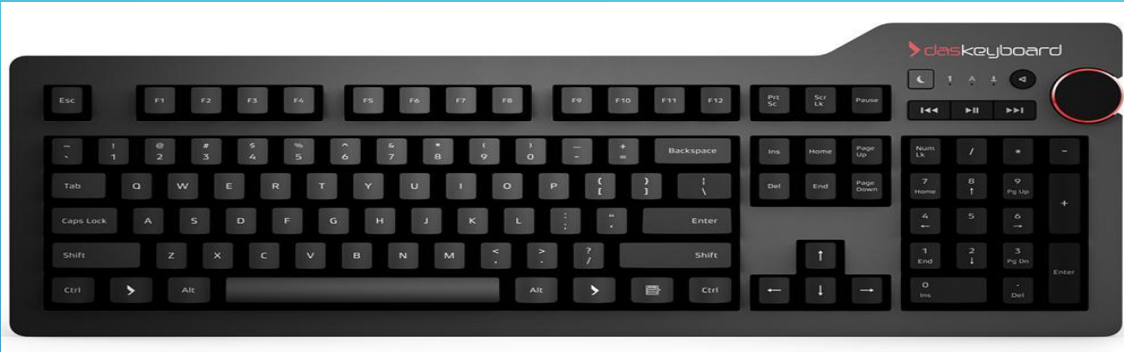
- Computer Hardware

refers to the physical equipment or components of an electronic data processing. It is also composed of input, output and storage and network devices.

Input Devices

- Input devices are the hardware devices which take information from user of the computer system, convert it into electrical signals and transmit it to the processor.

Keyboard



A computer hardware that used to input text characters and other commands.

Mouse

A device that used to point to and select items on your computer screen.



Image scanner

A Device that used to transfer images or text a computer there are special models the negatives or to scan book.



Web Camera

It is low resolution video camera used to provide visual input.



Microphone

Microphone is an input device to input sound that is then stored in a digital form. This is used to add sounds.



Output Devices

It is used to present information to the user from a computer. Output devices take data from the computer system and convert it to a form that can be read by humans.

Monitor

The most important difference between these monitor types is the technology used to create an image.

Cathode-ray (CRT)

The most common monitor type, red, green, blue electron beams. Most tv's used this technology



Liquid crystal display (LCD)

Is commonly used in laptops and some projectors. It consist of two polarizing filters with a liquid crystal solution.



Digital light processing (DLP)

Is a technology used in projectors, DLP projectors use a spinning color wheel with a microprocessor controlled array of mirrors called a digital micromirror device (DMD)



Printers And Fax Machines

Printers are output devices that create hard copies of computer files. Some printers specialize in part



Speakers and Headphones

Speakers and headphone are output devices for audio.



Storage Device

Alternatively referred to as **digital storage**, **storage**, **storage media**, or **storage medium**, a **storage device** is any hardware capable of holding information either temporarily or permanently. The picture shows an example of a drobo, an external secondary storage device.

Magnetic storage device

Magnetic storage is one of the most common types of storage used with computers.



Optical storage device

Another common storage device is this lasers and lights as its method of writing and reading data.



Flash memory

Has started to replace magnetic media, it is more efficient and reliable.

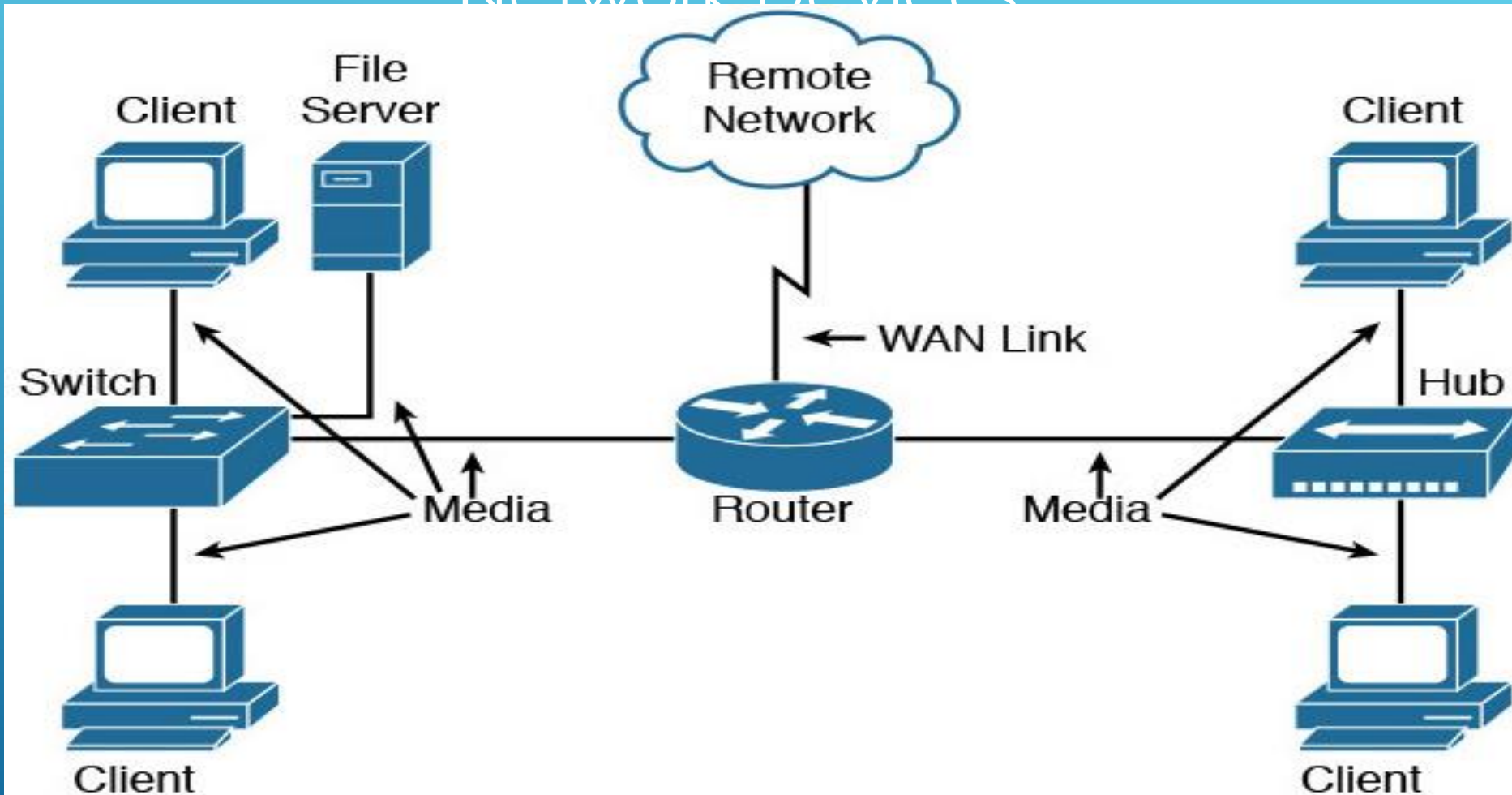


Paper storage

Early computers had no method of using any of the above technologies for storing information, these form is rarely used.



Network Devices



BRIDGE

It operates a data link layer. A bridge is a repeater, with add on functionality of filtering content by reading the MAC addresses of source and destination. It is used for interconnecting two LANS working on the same protocol.

Types of Bridge

Transparent Bridge

These are the bridge in which the stations are completely unaware of the bridges existence i.e.

Source Routing Bridge

Source route bridging is used on **token ring networks**, and is standardized in Section 9 of the **IEEE** 802.2 standard. The operation of the bridge is simpler (spanning tree protocol is not necessary) and much of the bridging functions are performed by the end systems, particularly the sources, giving rise to its name.

Switch

It is a multi port bridge with a buffer and a design that can boost can perform error, it divides collision domain of host.

Routers

In order for devices on the network to connect to the **Internet**, the **router** must be connected to a modem. Therefore, most **routers** have a specific Ethernet port that is designed to connect to the Ethernet port

Gateway

It is a passage to connect two networks together that may work upon different networking models.

Brouter

Also known as a bridging router which combines features of both bridge and router. It is working as a bridge, it can filter local area network traffic.